

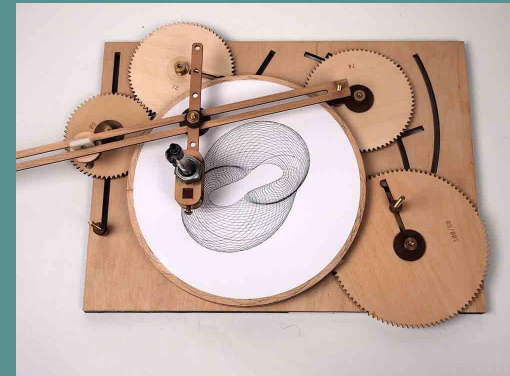
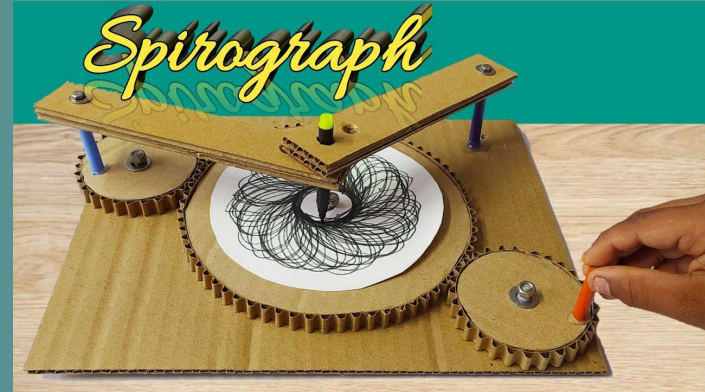
Spirograph

Team 4



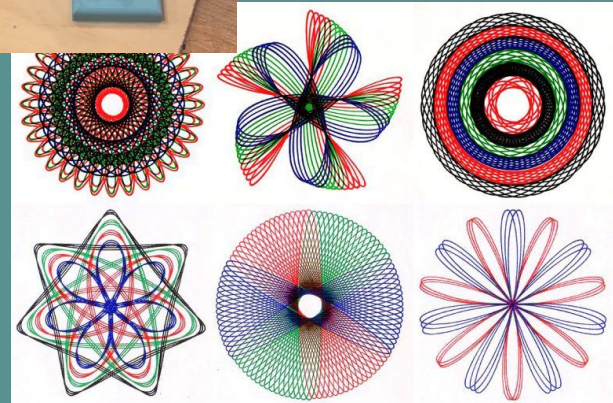
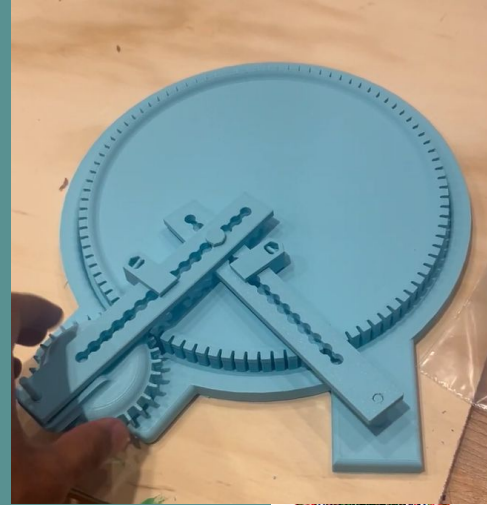
Inspiration & Motivation

- Inspired by classic Spirograph drawing toys that create geometric patterns using rotating gears
- Designed to explore how gears, links, and sliders work together to generate motion
- Demonstrates key principles of kinematics and mechanism design
- Converts a simple rotational input into complex drawing paths
- Produces circles, ellipses, and other custom patterns



Functionality

- Rotate the Small driving gear.
- Gear motion drives the linkage system
- Drawing point traces a path
- The combined gear and linkage motion generates smooth drawing paths
- Arm lengths and slider positions can be adjusted to create different patterns
- The toy can make circles, ellipses, and curves.



Analysis

Kutzbach Criterion:

$$m = 3(n-1) - 2j_1 - j_2$$

$$n = 5$$

$$j_1 = 6$$

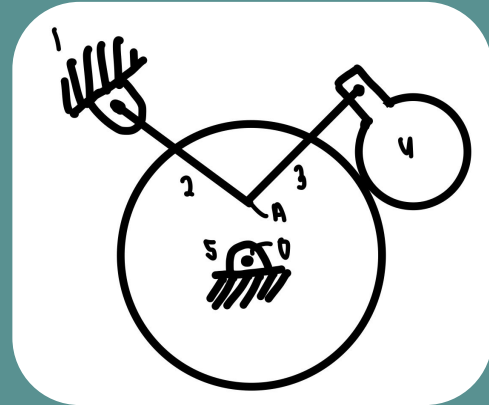
$$j_2 = 0$$

$$m = 0$$

Meshing gears are an exception to Kutzbach and subtract an additional degree of freedom.

Actual mobility:

$$m = 1$$



Simplified Model

Analysis + MATLAB

knowns:

$$R_1: 0.210 \text{ m} \quad \theta_1: 130$$

$$R_2: 0.105 \text{ m} \quad \theta_4: 45$$

$$R_3: 0.080 \text{ m}$$

$$R_4: 0.030 \text{ m}$$

position analysis:

$$\begin{cases} R_1 \cos(\theta_1) + R_4 \cos(\theta_4) + R_3 \cos(\theta_3) - R_2 \cos(\theta_2) = 0 \\ R_1 \sin(\theta_1) + R_4 \sin(\theta_4) + R_3 \sin(\theta_3) - R_2 \sin(\theta_2) = 0 \end{cases}$$

Jacobian:

$$\begin{bmatrix} -R_1 \sin(\theta_1) & R_4 \sin(\theta_4) \\ R_3 \cos(\theta_3) & -R_2 \cos(\theta_2) \end{bmatrix}$$

velocity analysis (leftover terms):

$$\begin{bmatrix} R_4 \sin(\theta_4) \omega_4 \\ -R_4 \cos(\theta_4) \omega_4 \end{bmatrix}$$

acceleration analysis (leftover terms):

$$\begin{bmatrix} R_4 \cos(\theta_4) \omega_4^2 + R_3 \cos(\theta_3) \theta_3^2 - R_2 \cos(\theta_2) \theta_2^2 \\ R_4 \sin(\theta_4) \omega_4^2 + R_3 \sin(\theta_3) \theta_3^2 - R_2 \sin(\theta_2) \theta_2^2 \end{bmatrix}$$

KC1:

$$0.1192$$

$$-0.3667$$

KC2:

$$-1.4054$$

$$0.9447$$

```
% vector loop equations
loopEqns = @(x) [
    r1*cos(th1) + r4*cos(th4) + r3*cos(x(1)) - r2*cos(x(2))
    r1*sin(th1) + r4*sin(th4) + r3*sin(x(1)) - r2*sin(x(2))
];

% solve for the unknowns
x = fsolve(loopEqns, x0);

% save values of unknowns
th3 = x(1);
th2 = x(2);

fprintf('\n----- Position Analysis Results ----- \n');
fprintf('th4 = %.6f rad\n', th4);
fprintf('th3 = %.6f rad\n', th3);
fprintf('th2 = %.6f rad\n', th2);
fprintf('----- \n');

%----- VELOCITY ANALYSIS -----
% Jacobian
J = [
    -R3*sin(th3) R2*sin(th2)
    R3*cos(th3) -R2*cos(th2)
];

% leftover terms
B1 = [
    R4*sin(th4)*om4
    -R4*cos(th4)*om4
];

% solve for unknown 1st order KCs
KC1s = J\B1;

% save values
th3p = KC1s(1);
th2p = KC1s(2);
```

```
xG2p = -r2/2 * sin(th2) * th2p;
yG2p = r2/2 * cos(th2) * th2p;
xG2pp = -r2/2 * (cos(th2)*th2p^2 + sin(th2)*th2pp);
yG2pp = r2/2 * (-sin(th2)*th2p^2 + cos(th2)*th2pp);
```

```
A2 = m2*(xG2p^2 + yG2p^2) + IG2*th2p^2;
B2 = m2*(xG2p*xG2pp + yG2p*yG2pp) + IG2*th2p*th2pp;
```

```
xG3p = -r2*sin(th2)*th2p - r3/2*sin(th3)*th3p;
yG3p = r2*cos(th2)*th2p + r3/2*cos(th3)*th3p;
xG3pp = -r2*(cos(th2)*th2p^2 + sin(th2)*th2pp) - r3/2*(cos(th3)*th3p^2 + sin(th3)*th3pp);
yG3pp = r2*(-sin(th2)*th2p^2 + cos(th2)*th2pp) + r3/2*(-sin(th3)*th3p^2 + cos(th3)*th3pp);
```

```
A3 = m3*(xG3p^2 + yG3p^2) + IG3*th3p^2;
B3 = m3*(xG3p*xG3pp + yG3p*yG3pp) + IG3*th3p*th3pp;
```

```
th4p = 1.0;
th4pp = 0.0;
A4 = IG4 * th4p^2;
B4 = IG4 * th4p * th4pp;
```

```
th5p = -1.0 / GR;
th5pp = 0.0;
A5 = IG5 * th5p^2;
B5 = IG5 * th5p * th5pp;
```

```
xA_p = -r2 * sin(th2) * th2p;
yA_p = r2 * cos(th2) * th2p;
vA_p = sqrt(xA_p^2 + yA_p^2);
```

```
dTdt = (A2 + A3 + A4 + A5)*om4*a14 + (B2 + B3 + B4 + B5)*om4^3;
```

```
dUgdt = (m2*g*yG2p + m3*g*yG3p)*om4;
dUdt = dUgdt;
```

Analysis

Assumptions/Inputs:

$\mu = 0.5N$	$m_2 = 0.035 \text{ kg}$	$I_{G2} = 5e^{-5} \text{ kgm}^2$
$F_N = 0.5N$	$m_3 = 0.025 \text{ kg}$	$I_{G3} = 3e^{-5} \text{ kgm}^2$
	$m_4 = 0.040 \text{ kg}$	$I_{G4} = 1e^{-5} \text{ kgm}^2$
	$m_5 = 0.150 \text{ kg}$	$I_{G5} = 4e^{-4} \text{ kgm}^2$

Force Analysis (done on Matlab):

$$T_{in} = \omega M_4 = dT/dt + dU/dt + dW_{dis}/dt$$

$$dT/dt = .000003 \text{ W}$$

$$dU/dt = 0 \text{ W}$$

$$dW_{dis}/dt = .003144 \text{ W}$$

Required Torque for Gear 4 Motor:

$$0.003147 \text{ N-m}$$

KNOWNs:
 $R_1: 0.210 \text{ m}$ $\theta_1: 130$
 $R_2: 0.105 \text{ m}$ $\theta_4: 45$
 $R_3: 0.080 \text{ m}$
 $R_4: 0.030 \text{ m}$

Position analysis:

$$\begin{bmatrix} r_1 \cos(\theta_1) + r_4 \cos(\theta_4) + r_3 \cos(\theta_3) - r_2 \cos(\theta_2) = 0 \\ r_1 \sin(\theta_1) + r_4 \sin(\theta_4) + r_3 \sin(\theta_3) - r_2 \sin(\theta_2) = 0 \end{bmatrix}$$

Solve for:

$$\begin{bmatrix} -r_3 \sin(\theta_3) + r_2 \sin(\theta_2) \\ r_3 \cos(\theta_3) - r_2 \cos(\theta_2) \end{bmatrix}$$

Velocity analysis (rotational terms):

$$\begin{bmatrix} r_4 \sin(\theta_4) \omega_4 \\ -r_4 \cos(\theta_4) \omega_4 \end{bmatrix}$$

Acceleration analysis (rotational terms):

$$\begin{bmatrix} r_4 \cos(\theta_4) \omega_4^2 + r_3 \cos(\theta_3) \theta_3^2 - r_2 \cos(\theta_2) \theta_2^2 \\ r_4 \sin(\theta_4) \omega_4^2 + r_3 \sin(\theta_3) \theta_3^2 - r_2 \sin(\theta_2) \theta_2^2 \end{bmatrix}$$

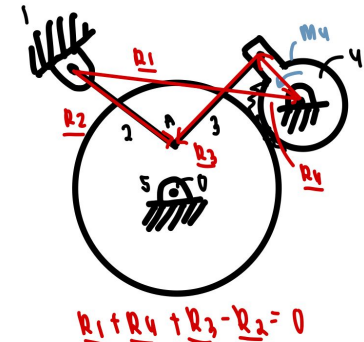
KC1:

$$\begin{bmatrix} 0.1192 \\ -0.3667 \end{bmatrix}$$

KC2:

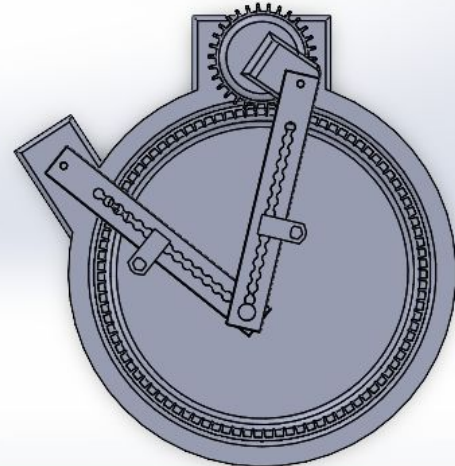
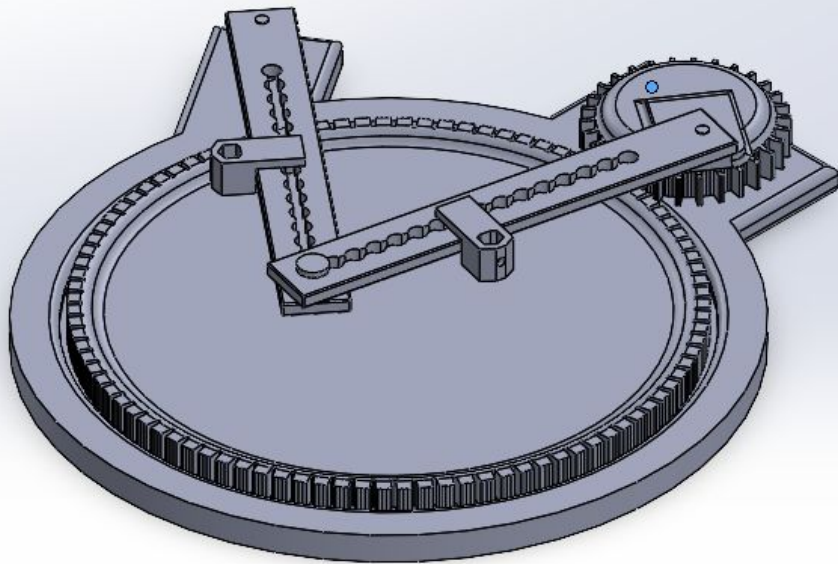
$$\begin{bmatrix} -1.4054 \\ 0.4497 \end{bmatrix}$$

KCs from Position/Velocity Analysis

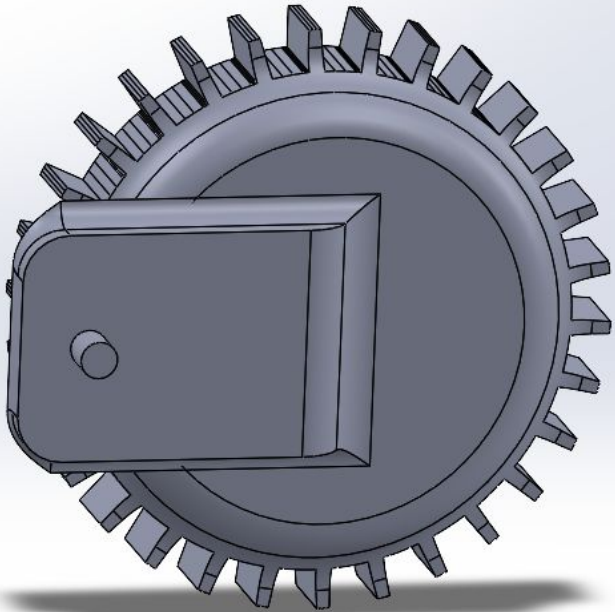


Simplified Model w/ Vector Loop

SolidWorks Design

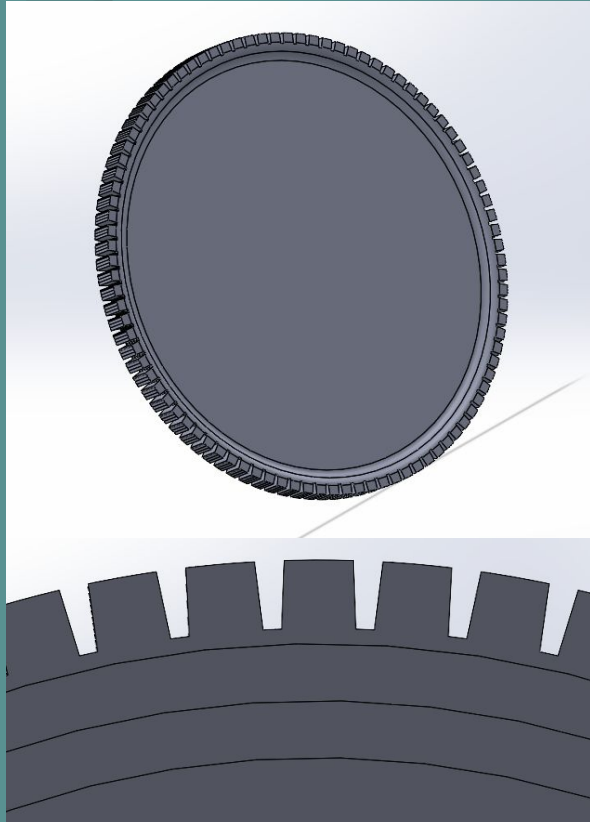


Driving Gear - Arm Extended



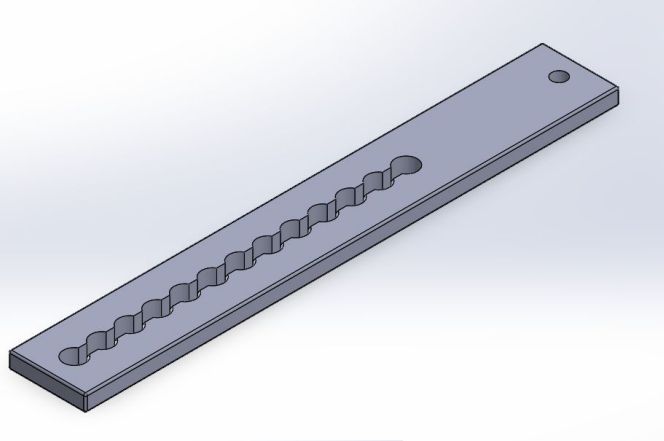
- 30 Teeth
- Pitch Diameter: 60mm
- Pressure Angle: 20 degrees
- Addendum: 2mm
- Dedendum: 2.5mm
- Face Width: 5mm
- Arm Width Offset: 10mm
- Extended Arm Pivot to become motor - future alternatives

Paper Tray/Ring Gear



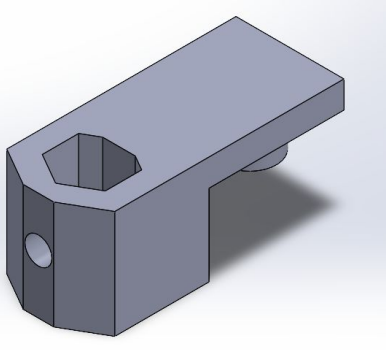
- 95 Teeth
- Pitch Diameter: 190mm
- Addendum Circle Diameter: 195mm
- Pressure Angle: 20 degrees
- Addendum: 2mm
- Dedendum: 2.5mm
- Face Width: 5mm
- Tooth Thickness: 4.55mm
- Speed Ratio: 3.167:1
- Paper Tray Depth: 5mm

Arm+Holder Design



- 150mm total length
- 3.5mm height
- 20mm width

- Moveable connection between arms
- Can move pen holder
- Holder height: 5mm



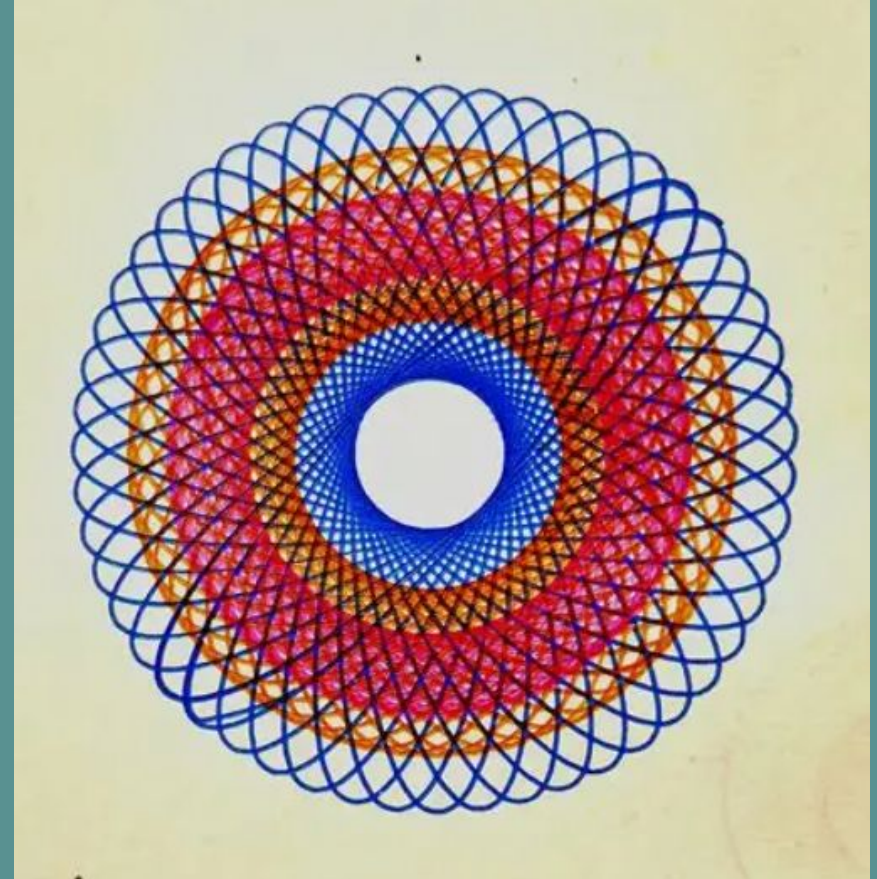
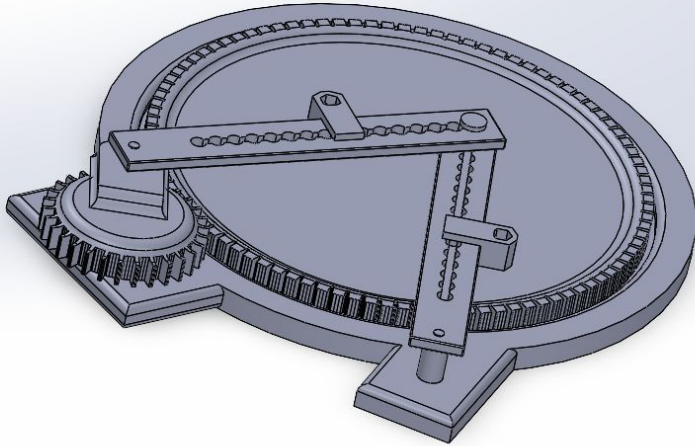


Difficulties in Designing

- Making sure the holders stayed within the bounds of the paper
 - lots of trial and error with different arm lengths
- Position of axis of motor gear assuming 3D printing tolerance
- Gear mates suck in SolidWorks

Desired Output

Adjustable arm pivot point and the location of the pen holder determines radius of the spirograph.



Where We Are/Improvements

- Smoothen gears
- Paper slides off
- Need bigger slots to hold the pencil/marker
- Attach Motor

